
Elementary Analysis

— *EYNTKA* Series —

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Frontmatter

Here (functions, numbers, sets, etc.)

Part I

Foundations of Single-Variable Analysis

This Part develops single-variable analysis on the real line: the construction and completeness of $\mathbb{R}$, sequences and series, continuity, differentiation, and the Riemann integral. It is the single-variable foundation the Calculus volume builds on.

1

sequences and series

HERE (Will look at other textbooks to see how this is implemented, but in many ways this is what analysis really starts with).

1.1 Sequences

2

Real Numbers

In this chapter, I will cover the construction of the real numbers

2.1 Construction Of The Real Numbers

The natural numbers can be thought as essentially being defined on the human intuition of counting¹. There are bigger sets of numbers, most of them generated by filling in gaps.

But what does it mean to fill in gaps? When you were first introduced to arithmetics, you were introduced to the basic operations: $+$, $-$, $\times$, $\div$, $\sqrt{}$, $\ln$.

However, Not all these operations will always return a value: in $\mathbb{N}$, there is no solution to the equation $3 - 5$, since $-2 \notin \mathbb{N}$ ². The integers extend the naturals making them closed under subtraction, that is

$$\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z} \text{ such that } a + b = 0$$

Usually, b is denoted $-a$, so $a + (-a) = 0$. Conventionally, this is shorthanded to $a - a = 0$.

Similarly, $\mathbb{Q}$ can extend $\mathbb{Z}$ by making them closed under division, that is

$$\forall a \in \mathbb{Q}, \exists b \in \mathbb{Q} \text{ such that } ab = 1$$

Usually, b is denoted a^{-1} or $\frac{1}{a}$, so $aa^{-1} = 1$ or $\frac{a}{a} = 1$.

One might think this covers all possible “gaps” which existed before. However, we

¹possible that other animals have the same intuition, like humming birds or horses, but right now, I'll stick to human's since as far as we know, humans are the only creatures that can conceptualize numbers beyond 20

²Or more precicely, no $a \in \mathbb{N}$ can be used as a solution to $3 - 5 = a$

Proposition 2.1.1: $\sqrt{2}$ is irrational

There does not exist a $a \in \mathbb{Q}$ such that $a = \sqrt{2}$

Proof:

For the sake of contradiction, let's assume that $\sqrt{2}$ is rational. Let $a = \frac{p}{q}$ ($p, q \in \mathbb{Z}$) where $\gcd(p, q) = 1$. If $\gcd(p, q) \neq 1$, simply divide $\frac{p}{q}$ by $\gcd(p, q)$. Since we're assuming $\sqrt{2}$ is rational:

$$\sqrt{2} = \frac{p}{q}$$

thus

$$2 = \frac{p^2}{q^2} \Rightarrow 2q^2 = p^2$$

From here, we'll take 2 cases:

p is odd: If p is even, then p^2 is odd. However, no matter what q is, $2q^2$ is even. Thus p cannot be odd

p is even: since $\gcd(p, q) = 1$, q cannot be even, since then $\gcd(p, q) = 2$. Thus, q is odd. Since p is even, it's divisible by 2. Thus p^2 is divisible by 4. However, $2q^2$ is only divisible by 2, meaning that p^2 and $2q^2$ must be different numbers, and hence p cannot be even.

However, all numbers in $\mathbb{Z}$ are either even or odd, thus we reached a contradiction.

Thus, $\mathbb{Q}$ is not closed under $\sqrt{\cdot}$. Furthermore, numbers like φ (the golden ration) is also not rational. We can fill in all these gaps by saying that any polynomial equation must have a solution. Thus

$$x^2 + 1 = 0 \Rightarrow x = \sqrt{2} \quad x^2 - x - 1 = 0 \Rightarrow x = \varphi$$

However, this too has it's limits. In 1873, Euler proved that e cannot be the solution of a polynomial equation. Later, it was shown that $\pi, \ln(2)$, and e^π each cannot be the solution to a polynomial equation. So, a further class of numbers was invented: If we can devise an algorithm to determine a number, the number is *computable*. Essentially, all numbers you can think about are computable (e, π , etc.). However, there do actually exist *uncomputable* numbers, numbers that we know exist, but we cannot devise an algorithm to define it! An example of this would be the Chaitin number. So, we need to find a way of defining an even bigger category of numbers, one with hopefully no gaps.

The way we'll do this is by defining a larger set, based off of $\mathbb{Q}$, which will represent all the gaps:

Definition 2.1.2: Dedekind Cut

A **cut** in $\mathbb{Q}$ is a pair of subsets A, B of $\mathbb{Q}$ st

1. $A \cup B = \mathbb{Q}$, $A \neq \emptyset$, $B \neq \emptyset$, $A \cap B = \emptyset$
2. if $a \in A$ and $b \in B$, then $a < b$
3. A contains no largest element

Let $x = A|B$ denote the [dedekind] cut

Example 2.1.3: here

1. $\{r \in \mathbb{Q} | r < 0\} | \{r \in \mathbb{Q} : r \geq 0\}$. Notice how this split kind of looks like $(-\infty, 0) | [0, \infty)$
2. $\{r \in \mathbb{Q} | r \leq 0 \text{ or } r^2 < 2\} | \{r \in \mathbb{Q} : r^2 \geq 2\}$. Notice how this split kind of looks like $(-\infty, \sqrt{2}) | [\sqrt{2}, \infty)$
3. Note that $\pm\infty$ is not a Dedekind cut, since that would imply $\emptyset | \mathbb{Q}$, or $\mathbb{Q} | \emptyset$, but A and B cannot be empty.

Note that in most constructions, A is the more interesting set, and B contains the “rest” of the rationals.

We will give a name to this set of cuts:

Definition 2.1.4: Real Numbers

The set of all Dedekind cuts of $\mathbb{Q}$ is called the **real numbers**, denoted $\mathbb{R}$. A **real number** is a cut in $\mathbb{Q}$.

We will justify calling this set a set of numbers soon. We'll first show that this indeed contains the rationals, and that it no longer contains “gaps” (meaning it contains all previous numbers we talked about, and more).

Notice that every rational number can be constructed using a Dedekind cut:

2.1.5 or any $q \in \mathbb{Q}$, $A|B = \{r \in \mathbb{Q} | r < q\} | \{r \in \mathbb{Q} | r \geq q\}$ is a Dedekind cut

Thus, we can identify every $q \in \mathbb{Q}$ with a $q^* \in \mathbb{R}$. In this way, we have that $\mathbb{Q}^* \subseteq \mathbb{R}$.

The next question is are there still gaps? Is there a place we can cut $\mathbb{R}$ to split it? This would mean we would need $A|B$, where $A \cup B = \mathbb{R}$, $A \cap B = \emptyset$, and $A, B \neq \emptyset$. The way we'll solve this is by first translating the order relation given to us in the definition of a dedekind cut to the real numbers:

Definition 2.1.6: defining an order relation on $\mathbb{R}$

If $x = A|B$ and $y = C|D$ are cuts such that $A \subseteq C$, then x is **less than or equal to** y and we write $y \leq x$. If $A \subseteq C$ and $A \neq C$, then x is **less than** y and we write $x < y$.

It can easily be proven that this is indeed an order relation.

2.1.7 Note The definition is essentially only reliant on the first set in each Dedekind cut. This further emphasizes that it's the main set we care about. The 2nd set can really be thought of as the "remainder" set

With the notion of an order, we can now also bound sets/numbers:

Definition 2.1.8: Upper And Lower Bound

An element $M \in \mathbb{R}$ is an **upper bound** for a set $S \subseteq \mathbb{R}$ (or S is bounded from above by M) if each $s \in S$ satisfies

$$s \leq M$$

Similarly for **lower bound**.

Definition 2.1.9: Least Upper Bound and Greatest Lower Bound

A *least upper bound* (or supremum) is an upper bound less than any other upper bound for S . We denote the least upper bound for S as $\sup(S)$.

Example 2.1.10: example 3

-1 is the least upper bound for the negative integers, while 4 is an upper bound.

Not all sets have a least upper bound. For example, in $\mathbb{Q}$, there is no least upper bound for $\sqrt{2}$ (we will show this when we prove $\mathbb{Q}$ is dense in $\mathbb{R}$). A set that contains all its least upper bounds is called a *complete* set. This is the key property that $\mathbb{R}$ has:

Theorem 2.1.11: $\mathbb{R}$ Is Complete

The set $\mathbb{R}$ is **complete** in the sense that it satisfies the least upper bound property: If S is a nonempty subset of $\mathbb{R}$ and is bounded above in $\mathbb{R}$, then there exists a least upper bound for S

Proof:

Let $C \subseteq \mathbb{R}$ be a nonempty subset of $\mathbb{R}$ that is bounded from above (so $\forall X|Y \in C, X|Y < A|B$ for some $A|B \in \mathbb{R}$, or written in shorthand $\forall x \in C, x < y$ for some $y \in \mathbb{R}$). Define the following Dedekind cut:

$$\begin{aligned} C &= \{a \in \mathbb{Q} \mid A|B \in C \text{ such that } a \in A\} \\ D &= \text{rest of } \mathbb{Q} \end{aligned}$$

Let $z = C|D$. It is easy to check that z is indeed a cut. We need to show that $z \leq y$ for any upper bound y . Indeed, let $y = C'|D'$ be any upper-bound for z . Then $C \subseteq C'$. Thus, $z \leq y$, showing that z is indeed the least upper bound

With this property, we can return to trying to make a Dedekind cut of the real numbers. Let's say $A|B$ is a cut of the real numbers. By definition, A contains no largest element. Furthermore, each

$b \in \mathcal{B}$ is an upper-bound for $\mathcal{A}$. Therefore, there exists a $y = \sup(\mathcal{A})$. Therefore, every Dedekind cut in $\mathbb{R}$ occurs at a real number!

2.1.1 Thinking of $\mathbb{R}$ as numbers

I mentioned earlier that $\mathbb{R}$ can be thought of as numbers. But what does it mean to be a number? The term “number” is actually an archaic term nowadays. Structures labeled as “numbers” are those which were historically labeled as numbers, since they were the ones studied the most. However, different types of numbers can have incredibly different properties. For example $\mathbb{N}$ and $\mathbb{R}$ are both labeled as numbers, but $\mathbb{N}$ and $\mathbb{R}$ are very different from one another! In mathematics, we have expanded this notion greatly in the domain of *Algebra*. For our purposes, we want $\mathbb{R}$ to have all the operations we have learnt in arithmetics:

$$+ \quad - \quad \times \quad \text{div}$$

and that they have all the properties we have grown accustomed to:

1. $a + b = b + a$
2. $a + (b + c) = (a + b) + c$
3. $\exists 0 \in \mathbb{R}$ such that for every $a \in \mathbb{R}$, $0 + a = a$
4. $\forall a \in \mathbb{R}$, $\exists -a \in \mathbb{R}$ such that $a - a = 0$
5. $a \cdot b = b \cdot a$
6. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
7. $\exists 1 \in \mathbb{R}$ such that for every $a \in \mathbb{R}$, $1 \cdot a = a$
8. $\forall a \in \mathbb{R}$, $\exists a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$
9. $a(b + c) = ab + ac$
10. if $x \leq y$ then $x + z \leq y + z$ for any $z \in \mathbb{R}$
11. $[\mathbb{R} \text{ is complete}]$

We have just shown the 10th and 11th property, so we'll go over some of the others. I say some, because it becomes quite tedious to prove all of them. They are all doable, but we will quickly start accumulating many very similar cases.

To define $+$ such that $x + y = E|F \in \mathbb{R}$ works as we are familiar with, we define:

$$\begin{aligned} E &= \{r \in \mathbb{Q} \mid \text{for some } a \in A \text{ and some } c \in C, \text{ we have } r = a + c\} \\ F &= \text{rest of } \mathbb{Q} \end{aligned}$$

This operation is naturally well-defined. By definition, we also have that $x + y = y + x$.

With this same idea, we can define the zero cut 0^* , and show that $0^* + x = x$ for every $x \in \mathbb{R}$. We can also show associativity in the same way

To define $\times$, it becomes a bit harder, since there are cases when $x > 0^*$ or $x < 0^*$. However, besides that, it's the same principle.

Also how we say *the* real numbers because this construction is unique, making the word “number” a good use, since we usually think of numbers being the usual thing.

2.1.2 What about further extending $\mathbb{R}$?

comment on the hyperreals. They contain the reals. They do not form a metric space. they are an ordered topology though. Also no unique hyperreals

2.2 Cauchy and Convergent Sequences

We will take a second look at completeness. We have shown that number systems smaller than the real numbers leave some gaps. We tried showing those by what seemed like hand-wavy methods. I gave examples like golden ration, then π , then the chaitin number, to keep showing there are gaps, but that's an inconsistent way of finding gaps, especially for more general sets which we might want to prove there is not gap. Is there one way to test, one way to find gaps, that a system must satisfy in order to test to see whether a set has gaps? Furthermore, can we think of the concept of completeness beyond the idea of numbers? Are there more general spaces for which the concept of completeness applies? This is what we cover in this section.

The main idea that was used to capture this idea is that if we try to approach any potential “hole” with a sequence of numbers, that number *must* exist in our set. For example,

$$f : \mathbb{N} \rightarrow \mathbb{Q} \quad x_n \in \left(\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}\right) \cap \mathbb{Q}$$

where x_n is some rational number inside the ever smaller open interval. Then this sequence approach $\sqrt{2}$, however $\sqrt{2} \notin \mathbb{Q}$, and it doesn't technically *converge* since for a sequence to converge, it must converge *to* a number. But in $\mathbb{Q}$, $\sqrt{2}$ is *not* a number, so it can't converge in it! We thus define what it means for a sequence to get arbitrarily close to “itself” as $n \rightarrow \infty$.³

Notice further that this new criterion for finding holes relies on sequences, in particular sequences that converge to one point. This might actually mean that completeness is a notion that can be defined on spaces where there is a single point a sequence can approach. There are actually a couple more conditions that are also required; when taken together they define something called a uniform space. It's in the framework of uniform spaces that we can define something called *uniform properties*, which are properties that are *global* to the space. Completeness is a uniform property, since the entire space is uniform. If you heard of uniform continuity, this also is a uniform property (we will cover uniform continuity in section ref:HERE).

However, there is little advantage to build-up the formal construction of uniform spaces, since we would only use very briefly here to how completeness can be extended in full generality, but then not use it again (maybe forever in your mathematical career). Instead, we will prove how a subset of

³I will quickly point out that though I used a computable function to find the hole, and basically all examples one can think of will also find be computable, you can have uncomputable Cauchy functions, meaning numbers like the Chitin number are also considered

uniform spaces which we use much more often can be called complete. That subset is the class of metric spaces!⁴.

To start off with, let's formally define cauchy sequences in a metric space:

Definition 2.2.1: Cauchy Sequence

If (X, d) is a metric space, we say that a sequence (x_n) is a Cauchy sequence if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_n, x_m) < \epsilon$ whenever $n, m > N$.

Intuitively, this means that after we get far enough in the sequence, any two points will be arbitrarily close to each other. Cauchy sequences are more general than convergent sequences:

Example 2.2.2: ample 1

Take $X = \mathbb{Q}$, with the standard metric. Take the sequence:

$$x_1 = 1, x_{n+1} = 1 + \frac{1}{1 + x_n}.$$

Notice that $d(x_n, x_m) < \epsilon$, and therefor it's Cauchy, but $d(x_n, x) < \epsilon$ is not true because it's a infinite continued fraction, that will equal to the golden ratio φ . Since $\varphi \notin \mathbb{Q}$, it doesn't converge to any x . Another example taking advantage of different metrics is:

$$X = \mathbb{R}, d(x, y) = |e^x - e^y|, (x_n) = -n$$

However, all convergent sequences are cauchy:

Proposition 2.2.3: convergent sequence $\Rightarrow$ Cauchy sequence

if (x_n) is a convergent sequence in (X, d) , then (x_n) is Cauchy. (Note that being Cauchy does not mean it converges.)

Proof. Suppose $(x_n) \rightarrow x$ and fix an $\epsilon > 0$. Choose an $N \in \mathbb{N}$ such that $d(x_n, x) < \frac{\epsilon}{2}$ if $n > N$. This same N shows that (x_n) is Cauchy, for if $m, n > N$ then

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Which suffices to prove it. □

We can ask when the converse holds? That is, when is it guaranteed that when approaching a point, that point will exist? This essentially is the idea of completeness we formulated earlier said in a different way, since it says any time we approach what we might suspect is a gap, we can guarantee there is an element there! To formalize this intuition:

⁴Historically, completeness of metrics was studied first. Later, when the concept was abstracted, uniform spaces were defined in terms of pseudometrics, and were found to be the natural generalization for studying completeness, but they are rarely used for current lack of utility beyond their use in generalization

Theorem 2.2.4: $\mathbb{R}$ Is Complete Using Cauchy Sequences

$\mathbb{R}$ is *complete* with respect to Cauchy sequences in the sense that if (a_n) is a sequence of real numbers which obeys a Cauchy condition then it converges to a limit in $\mathbb{R}$

Proof:

P.18-19. Much shorter proof than in 257. Basically showing first that it's bounded, then appropriately constructing a set that keeps bounded the sequence from above as $n \rightarrow \infty$, then take the sup of that set. Then prove that the Cauchy sequence converges using the sup value.

This will become the de-facto way one thinks about completeness, since it generalizes the completeness proof presented in 2.1.11 for arbitrary metric spaces.

Notice though that it works for metric spaces, and that the proof relied on the metric at hand. You can actually then make Cauchy sequences in spaces that seem to be complete, but are not (ex. look at the e^x metric on $[0, a]$ we defined earlier). You can even do more than that: you can have different metrics and construct different number systems! For example, if we let p be a prime number, then let $d(x, y) = |x - y|_p$ where $|a|_p = \frac{1}{p^k}$ where p^k is the largest power of p that divides a , then this will define a different metric that is *not* the Euclidean metric. This will in fact induce a different order on $\mathbb{Q}$ than the usual order, that is, it will no longer be that

$$1/2 \leq 1/5$$

but HERE. This links us back to our results from earlier in this chapter, where the construction of $\mathbb{R}$ was intimately linked to the order of $\mathbb{Q}$.

Since it's a metric space, it will define Cauchy sequences, and so we can complete the space too! The completion of $\mathbb{Q}$ through this metric is called labeled $\mathbb{Q}_p$ and is called the p -adic numbers!

I remember hearing that the real numbers are the unique complete ordered field, but the p -adics have the p -adic order, so I don't know how to combine the two.

2.3 Further properties of $\mathbb{R}$

In this section, we'll introduce a couple more properties the real numbers grant us.

Density

In topology, Given a space X , a set S is *dense* if $\overline{S} = X$. We can topologically prove that $\overline{\mathbb{Q}} = \mathbb{R}$. In real analysis, we would always be taking subsets $S \subseteq \mathbb{R}$ such that $\overline{S} = \mathbb{R}$. Density in $\mathbb{R}$ has a special property:

Proposition 2.3.1: Density In $\mathbb{R}$

A subset $S \subseteq \mathbb{R}$ is said to be dense in $\mathbb{R}$ if between any two real numbers, there exists an element of S . That is, if for any real number a, b such that $a < b$, then we have $S \cap (a, b) \neq \emptyset$

Proof:

Exercise.

Example 2.3.2: density

The most famous such examples is that $\mathbb{Q}$ is dense in $\mathbb{R}$. A problem with doing.

Archimedean principle**Definition 2.3.3: Archimedean Property**

Given an ordered space X , if for every $x \in X$, there is an integer n that is greater than x , then X has the *archimedean property*

Example 2.3.4: Archimedeana

$\mathbb{R}$ has the archimedean property. For any $x \in \mathbb{R}$, there exists a $n = \lceil x \rceil + 1$

 ϵ -principle

Finally, we'll label a fact that is very common, and you might've implicitly internalized it already doing continuity, convergent sequence, and limit proofs:

Theorem 2.3.5: ϵ Principle

If a, b are real numbers and if for each $\epsilon > 0$, we have $a \leq b + \epsilon$, then $a \leq b$. If for each $\epsilon > 0$, $|x - y| \leq \epsilon$, then $x = y$

Proof:

easy to prove.

3

Differentiation

here

Integration

here

4.1 Differentiation Under the Integral Sign

A fundamental question in analysis is identifying the conditions under which we can commute the integral and derivative operators. This is formalized by the Leibniz Integral Rule.

Theorem 4.1.1: Leibniz Integral Rule (Constant Limits)

Let $f(x, t)$ be a function defined for $x \in [a, b]$ and $t \in [c, d]$. Suppose that:

1. $f(x, t)$ is continuous on the rectangle $[a, b] \times [c, d]$.
2. The partial derivative $\frac{\partial f}{\partial t}$ exists and is continuous on $[a, b] \times [c, d]$.

Then, $F(t) = \int_a^b f(x, t) dx$ is differentiable on (c, d) and:

$$\frac{d}{dt} \left(\int_a^b f(x, t) dx \right) = \int_a^b \frac{\partial}{\partial t} f(x, t) dx$$

Proof:

Let $F(t) = \int_a^b f(x, t) dx$. By the definition of the derivative:

$$F'(t) = \lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h} = \lim_{h \rightarrow 0} \frac{\int_a^b f(x, t+h) dx - \int_a^b f(x, t) dx}{h}$$

By the linearity of the integral, we can combine the terms:

$$F'(t) = \lim_{h \rightarrow 0} \int_a^b \frac{f(x, t+h) - f(x, t)}{h} dx$$

We now fix x and consider $f(x, t)$ strictly as a function of t . By the Mean Value Theorem, for any fixed x , there exists a value $c_{x,h}$ strictly between t and $t+h$ such that:

$$\frac{f(x, t+h) - f(x, t)}{h} = \frac{\partial f}{\partial t}(x, c_{x,h})$$

We substitute this back into our expression. We wish to show that the limit of the integral approaches $\int \frac{\partial f}{\partial t}$. To do so, we analyze the difference:

$$\left| \int_a^b \frac{\partial f}{\partial t}(x, c_{x,h}) dx - \int_a^b \frac{\partial f}{\partial t}(x, t) dx \right|$$

Using the inequality $|\int g| \leq \int |g|$:

$$\leq \int_a^b \left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x, t) \right| dx$$

Here we rely on the compactness of the domain. Since $\frac{\partial f}{\partial t}$ is continuous on the closed, bounded rectangle $[a, b] \times [c, d]$, it is *uniformly continuous*.

This implies that for any $\epsilon > 0$, there exists a $\delta > 0$ such that if the distance between two points is less than δ , the difference in function values is less than $\frac{\epsilon}{b-a}$.

Since $c_{x,h}$ is between t and $t+h$, we know $|c_{x,h} - t| < |h|$. If we choose $|h| < \delta$, then for all $x \in [a, b]$:

$$\left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x, t) \right| < \frac{\epsilon}{b-a}$$

Substituting this bound back into the integral:

$$\int_a^b \left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x, t) \right| dx < \int_a^b \frac{\epsilon}{b-a} dx = (b-a) \cdot \frac{\epsilon}{b-a} = \epsilon$$

Thus, as $h \rightarrow 0$, the difference goes to 0, proving the equality.

4.1.2 Note If the limits of integration also depend on t , i.e., $a(t)$ and $b(t)$, and are differentiable, the general Leibniz rule applies:

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t) \cdot b'(t) - f(a(t), t) \cdot a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t} dx$$

Part II

Analysis on Metric and Function Spaces

This Part lifts the analysis of the first Part off the real line and onto general metric and function spaces: completeness and the contraction principle, compactness, uniform convergence, and the function-space theorems (Stone-Weierstrass, Arzela-Ascoli, Baire) that the later analysis books consume.

5

Topologies Through the Lens of Analysis

In this chapter, we will further elaborate on the standard topology of the real numbers given their special properties.

First, standard $\mathbb{R}$ is metrizable, so a reminder of the definition of metric from 327:

Definition 5.0.1: Metric

Let X be a set. A *metric* on X is the function

$$d : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

such that

1. $d(x, y) = d(y, x)$
2. $d(x, y) + d(y, z) \geq d(x, z)$
3. $d(x, y) = 0$ if and only if $x = y$

5.0.2 Note the fact that it is an *if and only if* is in fact very important. Without that you can say

$$d(x, y) = 0 \quad \forall x, y \in X$$

and that would be a metric, and you can go on to say it would define the indiscrete topology, but this was purposely left out because it would then not make all metric spaces Hausdorff (as we'll show soon)

Definition 5.0.3: Metric Topology

If d is a metric on X , we call the *metric topology* on X , to be the topology generated by the basis

$$\mathcal{B} = \{B_d(x, \epsilon) \mid x \in X, \epsilon > 0\}$$

Thus, standard $\mathbb{R}$ would have

$$d(x, y) = |x - y|$$

Definition 5.0.4: Convergence in Metric Space

A function $f : \mathbb{N} \rightarrow M$ is said to *converge* if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, d(x_n, p) < \epsilon$$

Corollary 5.0.5: Convergence of sequence and sub-sequence

A sequence converges if and only if every subsequence converges. Furthermore, they converge to the same point

Proof:

Simple proof

Definition 5.0.6: Sequential Continuity

A function $f : M \rightarrow N$ is *continuous* if it preserves sequential convergence: f sends convergent sequences in M to convergent sequences in N , limits being sent to limits. That is, for each sequence (p_n) in M which converges to a limit p in M , the image sequence $(f(p_n))$ converges to fp in N .

Corollary 5.0.7: Continuity $\Leftrightarrow$ sequential continuity in metric spaces

Let M be a metric space. Then a function is continuous if and only if it is sequentially continuous.

Proof:

here

Corollary 5.0.8: Epsilon-Delta Continuity

A function is continuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } d(x, t) < \delta \Rightarrow d(f(x), f(t)) < \epsilon$$

Proof:

here

Definition 5.0.9: Homeomorphism

A function is homeomorphic if it's bijective and bi-continuous

Definition 5.0.10: Closed Set In Metric Spaces

A set S is a **closed set** if it contains all its limit points

This definition constrained to real-analysis is interesting, since it mirrors the idea of closure in different domains of mathematics. In Algebra, an algebraically closed field is one for which every binary operation produces a number, in particular $\sqrt{\quad}$. Closed here gives a similar connotation.

Definition 5.0.11: Open Set In Metric Space

A set is an **open set** if for each $p \in S$ there exists an $r > 0$ st

$$d(p, q) < r \quad \Rightarrow \quad q \in S$$

We will use the following notation (where M is the metric space)

Note also the *neighborhood* will refer to *open neighborhoods*.

We also define a product metric:

Definition 5.0.12: Metric On Products Space

The following are metric we usually define on $M = X \times Y$:

$$\begin{aligned} d_E(m, m') &= \sqrt{d_X(x, x')^2 + d_Y(y, y')^2} \\ d_{max}(m, m') &= \max \{d_X(x, x'), d_Y(y, y')\} \\ d_{sum}(m, m') &= d_X(x, x') + d_Y(y, y') \end{aligned}$$

Proposition 5.0.13: order-relation between metrics

$$d_{max} \leq d_E \leq d_{sum} \leq 2d_{max}$$

Proof:

here

Then here there is all your knowledge about compactness

Then here there is nests of compactness If $A_1 \supset A_2 \cdots \supset A_n \supset A_{n+1} \supset \dots$ Then (A_n) is a nested sequence of sets. Its intersection is:

$$\bigcap_{i=1}^{\infty} A_n = \{p | \text{for each } n \text{ we have } p \in A_n\}$$

Theorem 5.0.14: intersection of nested compact sequence

The intersection of a nested sequence of compact nonempty sets is compact and non-empty

Proof:

here

The diameter of a nonempty set $S \subseteq \mathbb{R}^n$ is the supremum of the distances $d(x, y)$ between points of S

Corollary 5.0.15: Nested Compact Sequence to one-point

If in addition to being nested, nonempty, and compact, the sets A_n have diameter that tends to 0 as $n \rightarrow \infty$ then $A = \bigcap A_n$ is a single point

Proof:

p.83

Next is the concept of uniform continuity, which preserves global properties of spaces (ex. completeness, compactness, etc.)

Definition 5.0.16: Uniform Continuity

here

5.0.1 Clustering and Perfect Metric Space

Definition 5.0.17: Limit Point

here

Definition 5.0.18: Cluster Point

Let M be a metric space and let $p \in M$. Then p is a cluster point if for every $r > 0$

$$|B_r(p) \cap S| \geq \aleph_0$$

If $S \subseteq M$ is a set such that p is a cluster point around S , then we say S clusters at p .

Definition 5.0.19: Condensation Point

Let M be a metric space and let $p \in M$. Then p is a condensation point if for every $r > 0$

$$|B_r(p)| \geq \aleph_1$$

If $S \subseteq M$ is a set such that p is a condensation point around S , then we say S *condenses* at p .

Now, something linking metric spaces with completeness:

Definition 5.0.20: Perfect Metric

A metric space M is **perfect** if $M' = M$, i.e., each $p \in M$ is a cluster point of M . Recall that M clusters at p if each $M_r p$ is an infinite set.

Example 5.0.21: perfect metric

1. $[a, b]$ is perfect and
2. $\mathbb{Q}$ is perfect (meaning perfect $\nRightarrow$ complete).
3. $\mathbb{N} \subseteq \mathbb{R}$ is not perfect since none of its points are cluster points
4. If we generalize the notion of perfect to a more topological framework, a set is perfect if it's closed and has no isolated points (that is, x is an isolated point of S if there exists a neighborhood U such that $x \in U$ and $U \cap S = \{x\}$). Then $\mathbb{N}$ with the trivial topology has only perfect subsets, even though there are points not infinitely close to each other. This “infinitely close” fact comes from every ϵ -ball needing to not have isolated points (so it in a sense results from the fact that the co-domain of a metric is $\mathbb{R}$)

Theorem 5.0.22: Nonempty, Perfect, Complete Metric Space Is Uncountable

Every Nonempty, Perfect, Complete Metric Space Is Uncountable

Proof:

Using the fact that there must be at least an infinite number of points, we can at least create a countably long list of points. With this information try to construct a case to use Cantor's Diagonalization argument.

Corollary 5.0.23: Locally Uncountable

Every nonempty perfect complete metric space is everywhere uncountable in the sense that each r -neighborhood is uncountable.

Proof:

Take any ϵ -ball around any point x . take $\frac{\epsilon}{2}$. Taking it's closure will make it remain in the ϵ -ball. By 5.0.22, it is uncountable, and thus our ϵ -ball is uncountable

5.0.2 Arithmetical operations and continuity

Fun exercise: verify $+$, $-$, $\cdot$, div , and taking $1/x$ are all continuous (with div when the denominator is not zero). So is the $\max$, $\min$, and $|x|$ functions. So is scalar multiplication. This immediately implies polynomials are always continuous.

5.1 Compactness

In metric spaces, we can define the concept of sequential compactness:

Definition 5.1.1: Sequentially Compact

A subset $A \subseteq X$ of a metric space (X, d) is *sequentially compact* if for every sequence in A , there exists a convergent subsequence that converges in A

5.1.1 Compactness in compact metric spaces**Theorem 5.1.2: Generalized Heine-Borel Theorem**

A subset of a complete metric space is compact if and only if it is closed and totally bounded.

Proof:

p.104 Pugh

Sometimes, the following corollary is known as generalized Heine Borel. In $\mathbb{R}$:

$$\text{Compact} \Rightarrow \text{closed and bounded}$$

and for the converse in metric spaces:

$$\text{Compact} \Leftrightarrow \text{complete and totally bounded}$$

which is the result of his next corollary

Corollary 5.1.3: Compactness Of Metric Space (Generalized Heine-Borel)

A metric space is compact if and only if it is complete and totally bounded

Proof:
p.104

summary

[covering] compactness : every open cover has finite subcover

sequentially compact: every sequence has a convergent subsequence

In $\mathbb{R}$, Heine Borel provides:

$$\text{compact subset} \Leftrightarrow \text{closed and bounded subset}$$

In metric space:

$$\text{compact subset} \Rightarrow \text{closed and bounded subset} \quad \text{compact subset} \not\Leftrightarrow \text{closed and bounded subset}$$

Or, if we think about it in terms of the entire space:

$$\text{compact space} \Rightarrow \text{bounded space} \quad \text{compact space} \not\Leftrightarrow \text{bounded space}$$

counter example being $\mathbb{Q} \subseteq \mathbb{R}$

Example 5.1.4: Example

Take $[0, \frac{1}{\pi} - \frac{1}{n}] \cup [\frac{1}{\pi} + \frac{1}{n}, 1]$ for $n \geq 4$. This covers all of $\mathbb{Q} \cap [0, 1]$, but has no finite subcover. Note that these sets are open in the subspace topology, but not in $\mathbb{R}$.

However:

$$\text{compact space} \Leftrightarrow \text{complete and totally bounded}$$

and so in complete metric space:

$$\text{compact space} \Leftrightarrow \text{totally bounded}$$

and using theorem ref:HERE, we get:

$$\text{compact subset} \Leftrightarrow \text{closed and totally bounded subset}$$

Note that $C^0([a, b], \mathbb{R})$ is complete (a proof in prop ref:HERE). However, the infinite-dimensionality of it means we need another condition for compactness: Using Arzela Ascoli, we get:

$$\text{compact subset} \Leftrightarrow \text{closed, uniformly bounded, and equicontinuous}$$

For “closed and bounded $\Leftrightarrow$ compact” in a metric space you need:

$$\text{complete, } \sigma\text{-compact, locally compact}$$

which we’ll never use in these notes, but it’s a fun trivial.

5.2 The Contraction Principle

Completeness has a powerful fixed-point consequence: on a complete metric space, any map that uniformly shrinks distances has exactly one fixed point, and it is reached by iterating the map from any starting point. This is the engine behind many existence-and-uniqueness theorems in analysis, from the inverse function theorem to the existence and uniqueness of solutions of ordinary differential equations.

Definition 5.2.1: Fixed Point

A *fixed point* of a map $f : X \rightarrow X$ is a point $p \in X$ with $f(p) = p$.

Definition 5.2.2: Contraction

Let (X, d) be a metric space. A map $f : X \rightarrow X$ is a *contraction* if there is a constant k with $0 \leq k < 1$ such that

$$d(f(x), f(y)) \leq k d(x, y) \quad \text{for all } x, y \in X.$$

Every contraction is Lipschitz, and hence uniformly continuous: given $\epsilon > 0$, the choice $\delta = \epsilon$ makes $d(x, y) < \delta$ force $d(f(x), f(y)) \leq k d(x, y) < \epsilon$.

Theorem 5.2.3: Banach Contraction Principle

Let (X, d) be a nonempty *complete* metric space and let $f : X \rightarrow X$ be a contraction with constant k . Then f has a unique fixed point $p \in X$, and for every $x_0 \in X$ the iterates defined by $x_{n+1} = f(x_n)$ converge to p , with

$$d(x_n, p) \leq \frac{k^n}{1 - k} d(x_0, x_1).$$

Proof:

Fix $x_0 \in X$ and set $x_{n+1} = f(x_n)$. Applying the contraction bound repeatedly,

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq k d(x_n, x_{n-1}) \leq \cdots \leq k^n d(x_1, x_0).$$

Hence for $m > n$, the triangle inequality and the geometric series give

$$d(x_m, x_n) \leq \sum_{i=n}^{m-1} d(x_{i+1}, x_i) \leq \left(\sum_{i=n}^{m-1} k^i \right) d(x_1, x_0) \leq \frac{k^n}{1 - k} d(x_1, x_0).$$

Since $0 \leq k < 1$, the right-hand side tends to 0 as $n \rightarrow \infty$, so (x_n) is a Cauchy sequence. As X is complete, $x_n \rightarrow p$ for some $p \in X$.

This limit is a fixed point: because f is continuous, $f(p) = f(\lim_n x_n) = \lim_n f(x_n) = \lim_n x_{n+1} = p$. Letting $m \rightarrow \infty$ in the bound above (the distance is continuous in each argument) yields the stated error estimate.

The fixed point is unique: if $f(p) = p$ and $f(q) = q$, then $d(p, q) = d(f(p), f(q)) \leq k d(p, q)$, so $(1 - k) d(p, q) \leq 0$. Since $1 - k > 0$, this forces $d(p, q) = 0$, that is, $p = q$.

5.3 Basic Function Spaces

Example 5.3.1: pointwise convergence

1. here
2. Note that it's possible that a pointwise converging sequence of functions “miss” points. Take

$$f_n(x) : (a, b) \rightarrow \mathbb{R} \quad f_n(x) = x^n$$

then $f_n \rightarrow 0$, the zero-function, since every point is indeed eventually converging to 0. However, it seems like there will be a point which is sort of “up there”. We need a strong notion to fix that

Definition 5.3.2: Uniform Convergence

Let M and N be arbitrary metric spaces. A sequence of function $f_n : M \rightarrow N$ *converges uniformly* to the limit $f : M \rightarrow N$ if for each $\epsilon > 0$ there is an N such that for all $n \geq N$, and all $x \in M$,

$$|f_n(x) - f(x)| < \epsilon$$

The function f is the *uniform limit* of the sequence (f_n) and we write

$$f_n \rightrightarrows f \quad \text{or} \quad \text{unif} \lim_{n \rightarrow \infty} f_n = f$$

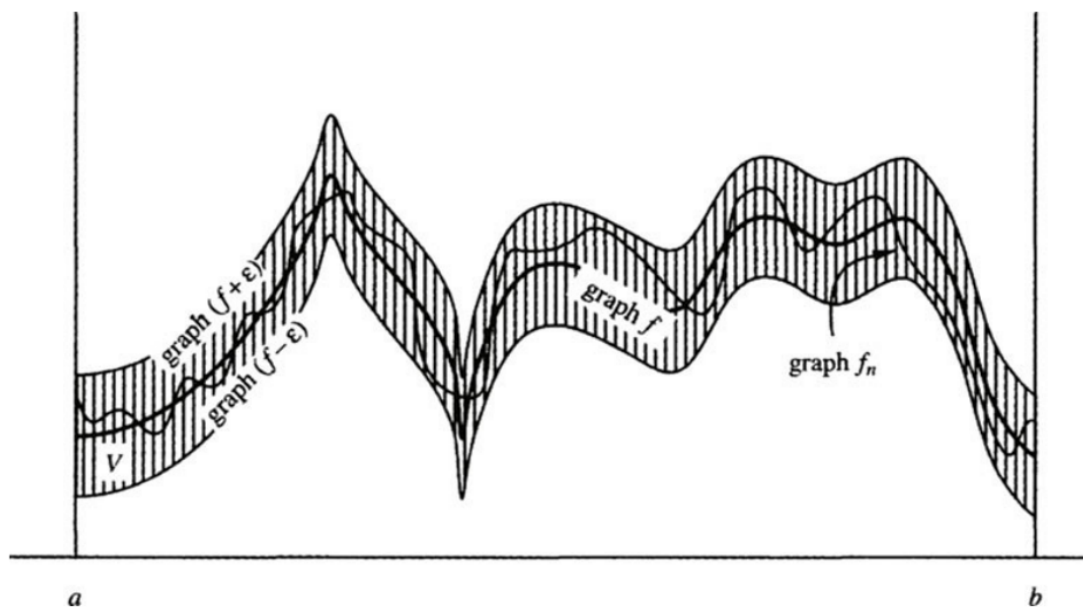


Figure 5.1: uniform convergence visually

Theorem 5.3.3: Uniform Convergence Preserve's Continuity

if $f_n \Rightarrow f$ and each f_n is continuous at x_0 then f is continuous at x_0

Proof:

Key is to use the triangle inequality. Details are omitted to be field. Once you choose any $\epsilon > 0$, you know for some $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\epsilon}{3}$$

Then you know for f_N at x_0 there is a $\delta > 0$ st

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}$$

and then you do the triangle inequality twice, to get split the first equation into 3 equations with f_N being part of it at least once, then use they're all less than $\frac{\epsilon}{3}$, and done

So, with $f_n(x) = x^n$, these function do *not* uniformly converge to 0, but to

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Note that the converse is not true: if at every point x_0 , f_n is continuous and f is continuous at x_0 , this does not imply (f_n) is uniformly convergent:

Example 5.3.4: growing steeple

This is called the growing steeple:

$$f_n(x) = \begin{cases} n^2x & \text{if } 0 \leq x \leq \frac{1}{n} \\ 2n - n^2x & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{if } \frac{2}{n} \leq x \leq 1 \end{cases}$$

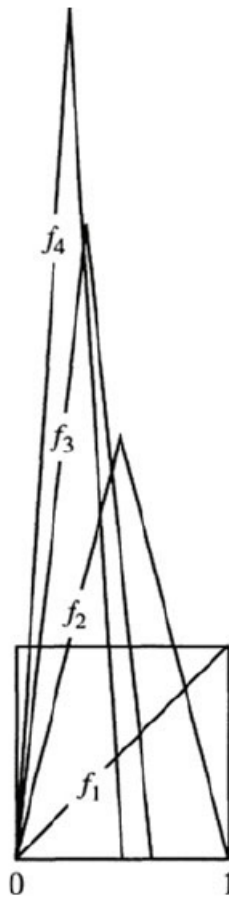


Figure 5.2: growing steeple function

and if you don't like that it goes to infinity, multiply it by $\frac{1}{n}$

Application to function spaces

Definition 5.3.5: Space Of Bounded Functions

Let $C_b([a, b], \mathbb{R})$ represent the set of all bounded functions from $f : [a, b] \rightarrow \mathbb{R}$

Then we can define the following norm on this space:

Definition 5.3.6: Sup Norm

Let $f \in C_b$. Then:

$$\|f\| = \sup \{|f(x)| | x \in [a, b]\}$$

you should check that this is indeed a norm, and that it will induce a metric as well, and hence a metric topology, and hence the idea of convergence.

In this metric space under this metric, the definition is the same as uniform convergence

Theorem 5.3.7: sup norm convergence if and only if uniformly convergence

Convergence with respect to the sup metric d is equivalent to uniform convergence

Proof:

The two definition's are in fact identical.

Since $d(f_n, f) \rightarrow 0$, then for every $x \in [a, b]$ the distnce is becomming smaller, so $|f(x) - f_n(x)|\epsilon$, and so $f_n \Rightarrow f$.

Theorem 5.3.8: C_b Is A Complete Metric Space

C_b is a complete metric space

Proof:

p.216. If you take any cauchy sequence of functions, then at every point, it's a cuachy sequence in $\mathbb{R}$. use the face that $\mathbb{R}$ is complete while using the sup-norm to complete the proof.

With this, we can show that the space of continuous bounded functions is complete too.

Definition 5.3.9: Continuous Bounded Funcitons

Let $C^0([a, b], \mathbb{R})$ denote the set of continuous functions from $f : [a, b] \rightarrow \mathbb{R}$. Note that since the domain is compact, the image is compact, and in $\mathbb{R}$ that implies it's bounded, so $C^0 \subset C_b$.

Corollary 5.3.10: C^0 Is A Compelte Metric Space

C^0 Is A Compelte Metric Space

Proof:

By theorem 5.3.3, every convergent sequence in C^0 lies in C^0 , thus C^0 contains it's limits, and thus is closed. so it's a closed subset of a compelte metric space. Thus, by HERE, it's complete

5.4 Application to other concepts

We'll now explore how uniform convergence works with derivatives, integrals, and series

Let

$$F_n = \sum_{i=1}^n f_i(x)$$

Then we can say that (F_n) converges uniformly just like with (f_n) since it's also a sequence of functions.

If we have

$$F_n = \sum_{i=1}^{\infty} f_i(x)$$

Then we can say it converges uniformly if the sequence of partial sums converges uniformly.

Also, we can say the series of absolute values $\sum |f_k(x)|$ converges then the series $\sum f_k$ converges *absolutely*

Theorem 5.4.1: Weierstrass M-Test

If $\sum M_k$ is a convergent series of constants and if $f_k \in C_b$ satisfies $\|f_k\| \leq M_k$ for all k then $\sum f_k$ converges uniformly and absolutely

Proof:

For n large enough, we can use the triangle inequality on the partial sums to expand out our series:

$$\begin{aligned} d(F_n, F_m) &\leq d(F_n, F_{n-1}) + \cdots + d(F_{m+1}, F_m) \\ &= \sum_{k=n}^m \|f_k\| \leq \sum_{k=n}^m M_k \end{aligned}$$

and thus, since $\sum M_k$ converges, then we can always use its epsilon for large enough m, n . This shows that (F_n) is Cauchy, and so since C_b is complete, it converges uniformly.

Theorem 5.4.2: Uniform Limits Through Integrals

The uniform limit of Riemann integrable functions is Riemann integrable, and the limit of the integrals is the integral of the limit

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx$$

Proof:

p.218

Note that since all bounded continuous functions are integrable, but not all integrable bounded functions are continuous, this means that the subset of C_b of functions for which this theorem applies is between

C_b and C_0 (since not all bounded functions are integrable. If $\mathcal{R}$ represent all integrable functions:

$$C_b \supset \mathcal{R} \supset C^0$$

Now, using the fact that a limit can pass through integrable functions, we can show that infinite sums of integrated functions is the same as the integral of the infinite sum of functions:

Theorem 5.4.3: Term By Term Integration Theorem

A uniformly convergent series of integrable functions $\sum f_k$ can be integrated term-by-term in the sense that

$$\int_a^b \sum_{k=0}^{\infty} f_k(x) dx = \sum_{k=0}^{\infty} \int_a^b f_k(x) dx$$

Proof:

Use the definition, then apply theorem 6.

Now, we move on to differentiable functions

Theorem 5.4.4: Uniform Limit Of Differentiable Functions Is A Differentiable Function

The uniform limit of a sequence of differentiable functions is a differentiable function provided that the sequence of derivatives also converges uniformly

Proof:

p.219-220

Assuming f' is continuous makes the proof easy enough using the fundamental theorem of calculus by splitting up $f_n(x) = f_n(a) + \int_a^x f'_n(t) dt$.

For f' not continuous, more to check. Check the book, it's not too bad to follow

This also looks cool: [here](#)

Note that this can fail if the derivative doesn't converge uniformly

Example 5.4.5: derivative not converge uniformly

Take

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

Then this series converges to $f(x) = |x|$, which is not differentiable! The derivative converges point-wise, but not uniformly.

Note that in complex analysis, this will not be the case! the uniform limit of a complex differentiable function is complex differentiable, and the sequence of derivatives will automatically be uniformly convergent to a limit!

Like before, infinite sums work well

Corollary 5.4.6: Series Of Infinite Sum And Uniform Convergence

A uniformly convergent series of differentiable functions can be differentiated term-by-term, provided that the derivative series converges uniformly,

$$\left(\sum_{k=0}^{\infty} f_k(x) \right)' = \sum_{k=0}^{\infty} f_k'(x)$$

Proof:

Using the definition of convergence for infinite series, apply theorem 5.4.4 on the partial sums.

5.5 Compactness and equicontinuity

We now want to apply the Heine-Borel theorem to C^0 , because checking something is closed and bounded is relatively easy, and for some reason, we want to be able to easily find convergent-subsequence of convergent sequences (will come back here once I figure out why TBD)

In C^0 , closed and bounded sets are usually not compact. Take

$$B = \{f \in C^0([0, 1], \mathbb{R}) \mid \|f\| \leq 1\}$$

that is, the set of all functions whose supremum over $[0, 1]$ is 1. This set is not closed: it contains convergent sequences that do not lie inside B , however the value it converges to is *not* continuous

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

This comes down to the fact that $f(x)$ is infinite dimensional: Heine-Borel will apply to finite-dimensional spaces!

To show that C^0 is an infinite dimensional vector-space over $\mathbb{R}$, I'll show that it has an infinite dimensional linearly independent set (I'll not be showing it's a basis here, but since a set exists, the basis must at least have this cardinality). Take the set of "smooth bump" functions, where f_0 starts its bump at 0 and ends at 1. Then $\{f_n \mid n \in \mathbb{Z}\}$ is a linearly independent set of continuous (even smooth) functions. Thus, the space must at least have a countably infinite dimension.

Now back to finding the appropriate conditions for the Heine-Borel property. To do this, we need a new condition on our functions:

Definition 5.5.1: [Uniform] Equicontinuous

A sequence of functions (f_n) in C^0 is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta, \forall n \in \mathbb{N} \Rightarrow |f_n(s) - f_n(t)| < \epsilon$$

The functions f_n are *equally continuous*.

Note that the δ depends on the ϵ , but *not* on the n I put [uniform] in the definition because there are definitions of equicontinuity which are not “uniform”, that is, you pick a point for which the conditions applies:

Definition 5.5.2: Point-Wise Equicontinuous

A sequence (f_n) in C^0 is *pointwise equicontinuous* if

$$\forall \epsilon > 0 \forall x \in [a, b], \exists \delta > 0 \text{ such that } |x - t| < \delta, n \in \mathbb{N}, |f_n(x) - f_n(t)| < \epsilon$$

From here on out, we will say equicontinuity instead of uniform equicontinuity.

Example 5.5.3: Example

1. The set of function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = \frac{x}{n}$ are equicontinuous. However, they are not uniformly continuous.
2. Take $f : [0, 1] \rightarrow \mathbb{R}$, $f_n(x) = n$. That is, a constant function. Then $\{f_n\}$ is an equicontinuous sequence. However $\{f_n\}$ is not a convergent sequence (and so not uniformly convergent).
3. let $\{f\}$ be a equicontinuous set with a single function. Then f is uniformly continuous by definition. In this sense, equicontinuity is intimately related to uniform continuity.

<+ +>

This definition can also be used to define an equicontinuous set:

Definition 5.5.4: Equicontinuous Set

The set $\mathcal{E} \subset C^0$ is equicontinuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta \text{ and } f \in \mathcal{E} \Rightarrow |f(s) - f(t)| < \epsilon$$

Essentially, if a function is continuous, then for all $\epsilon > 0$, there exists an open neighborhood U such that $d(f(x), f(y)) < \epsilon$ for all $y \in U$. Equicontinuity means that a single neighborhood U can be chosen that will work for all the functions f in the collection $\mathcal{E}$.

You might recall Bolzano-Weierstrass theorem:

Theorem 5.5.5: Bolzano-Weierstrass Theorem

Let $M = \mathbb{R}^n$. Then every bounded sequence has a convergent subsequence.

Proof:

Here is the sketch of the proof.

Since it's bounded, put it in a ball, and take the closure of the ball. Then it's closed. So it's compact. Take the cover: $B_1 = \{B_1(x) | x \in \mathbb{R}^n\}$. Then it has a finite sub-cover B'_1 . Then one of the ball must have infinitely many points. Let's say $b'_1 \in B'_1$. Then cover this ball with $B_{1/2}(x)$ balls, and repeat the same process. Keep doing this for every n . Then you can pick a

sub-sequence from each $B_{1/n}(x)$ which will converge.

We'll be proving the analogous of this with equicontinuous sequences in C^0 :

Theorem 5.5.6: Arzela-Ascoli Theorem

Let M be a compact metric space. Every uniformly bounded equicontinuous sequence of functions in $C^0(M, \mathbb{R})$ has a uniformly convergent subsequence

Proof:

p.226. Fun to follow!

Notice how this relates to compactness: if every subsequence of f_n converges, then the closure of $\{f_n | n \in \mathbb{N}\}$ is compact!

The last part of the proof also shows that you can weaken the condition a bit, and say that if you converge on a dense subset, then you're uniformly convergent.

Corollary 5.5.7: Arzela-Ascoli Propagation Theorem

Pointwise convergence of an equicontinuous sequence of functions on a dense subset of the domain *propagates* to uniform convergence on the whole domain

Proof:

last part of the proof for Ascoli

Another thing that I don't know where to put, but I think is neat to consider:

1. Let's say that you have a sequence of function where $f_n \rightarrow f$ pointwise. you also have sequences of functions $f_{n,m} \rightarrow f_n$, that is, every f_n has a sequence of function's converging to it. Then is there a subsequence of $f_{n,m}$ (for variable n, m) that converges to f ?

<++>

If your derivative is bounded, we can actually prove this with a much weaker claim: only 1 point must be convergent for the whole thing to be uniformly continuous!!

Corollary 5.5.8: Differentiable Convergent Subsequence

Assume that $f_n : [a, b] \rightarrow \mathbb{R}$ is a sequence of differentiable functions whose derivatives are uniformly bounded. If for one point x_0 , the sequence $(f_n(x_0))$ is bounded as $n \rightarrow \infty$, then the sequence (f_n) has a subsequence that converges uniformly on the whole interval $[a, b]$.

Proof:

Let M be the bound of $|f'_n(x)|$ for all $n \in \mathbb{N}$ and $x \in [a, b]$. Equicontinuity of (f_n) then follows from the MVT:

$$|s - t| < \delta \Rightarrow |f_n(s) - f_n(t)| = |f'_n(\alpha)| |s - t| \leq M\delta$$

and then set $\delta = \frac{\epsilon}{(M+1)}$. This shows that (f_n) is equicontinuous!

Now to show that the sequence is uniformly bounded, let C be a bound for $|f_n(x_0)|$ for all $n \in \mathbb{N}$. Then

$$\begin{aligned} |f_n(x)| &\leq |f_n(x) - f_n(x_0)| + |f_n(x_0)| \leq M|x - x_0| + C \\ &\leq M|b - a| + C \end{aligned}$$

showing that the sequence (f_n) is bounded in C^0 ! Since it's uniformly bounded and equicontinuous, by Arzela-Ascoli's theorem (5.5.6), $f(f_n)$ has a uniformly convergent subsequence!

As a quick aside, this theorem has two uses in other domains we will not discuss;

1. A sequence of solutions to a continuous ordinary differential equation in $\mathbb{R}^M$ has a subsequence that converges to a limit, and that limit is also a solution of the ODE
2. A sequence of complex analytic functions that converges pointwise, converges uniformly (on compact subsets of the domain of definition) and the limit is complex analytic.

However, we'll be using Arzela-Ascoli for Heine-Borel:

Theorem 5.5.9: Heine-Borel Theorem In A Function Space

A subset $\mathcal{E} \subseteq C^0$ is compact if and only if it is closed, bounded, and equicontinuous

Proof:

($\Rightarrow$) Since C^0 is a complete metric space, it is totally bounded (it is automatically closed since C^0 is closed). This means for some $\epsilon > 0$, we can cover $\mathcal{E}$ with finitely many balls of size $\epsilon/3$

$$\mathcal{C} = \{B_{\epsilon/3}(f_k) \mid \text{appropriate } f_k\}$$

Note that each f_k is uniformly continuous since the domain is compact. Thus, there exists a $\delta > 0$ st

$$|s - t| < \delta \Rightarrow |f_k(s) - f_k(t)| < \frac{\epsilon}{3}$$

Now, if $f \in \mathcal{E}$, then for some k we have $f \in B_{\epsilon/3}(f_k)$, and $|s - t| < \delta$ will give us

$$|f(s) - f(t)| \leq |f(s) - f_k(s)| + |f_k(s) - f_k(t)| + |f_k(t) - f(t)| < 3 \frac{\epsilon}{3} = \epsilon$$

showing $\mathcal{E}$ is equicontinuous.

($\Leftarrow$) Assume $\mathcal{E}$ is closed, bounded, and equicontinuous. Then if (f_n) is a sequence in $\mathcal{E}$, by Arzela-Ascoli's theorem, some subsequence converges uniformly to a limit. This limit lies in $\mathcal{E}$ since $\mathcal{E}$ is closed. Thus, by ref:HERE, $\mathcal{E}$ is compact

5.6 Uniform Approximation in C^0

We now move onto a really interesting part of analysis: Given a continuous but nondifferentiable function f (so, any function in C^0), we can “arbitrarily” approximate f with a smooth function

g ! Even better, this smooth function will be a polynomial!!

In order for this to be possible, it would mean that for any $f \in C^0$, there is a polynomial close to it. That is what we're about to prove:

Theorem 5.6.1: Weierstrass Approximation Theorem

The set of polynomials is dense in $C^0([a, b], \mathbb{R})$

Note that since the function is continuous on a compact set, it is uniformly continuous (a fact we'll use in the 2nd proof).

Decomposing what density means for a moment: for each $f \in C^0$, and each $\epsilon > 0$, there is a polynomial function $p(x)$ such that for all $x \in [a, b]$

$$|f(x) - p(x)| < \epsilon$$

meaning that we can arbitrarily approximate $f(x)$!!

Proof:

#1 The key idea to building $p(x)$ from f is to sample values of f and recombining them in a clever way.

Without loss of generality, let's say $[a, b] = [0, 1]$. For each $n \in \mathbb{N}$, consider

$$p_n(x) = \sum_{k=0}^n \binom{n}{k} c_k x^k (1-x)^{n-k}$$

where $c_k = f\left(\frac{k}{n}\right)$ and $\binom{n}{k}$ is the binomial coefficient. This polynomial is called the **Bernstein polynomial**. We'll show that (p_n) converges uniformly to f as $n \rightarrow \infty$, i.e., $(p_n) \rightrightarrows f$.

Proof:

#2 This proof uses convolution. to link the nice properties of polynomials with the less nice properties of general continuous functions.

Without loss of generality, let's let $[a, b] = [0, 1]$ (they are homeomorphic). Let $f \in C^0([0, 1], \mathbb{R})$. What we'll do is manipulate this functions to a nicer homeomorphic equivalent, and approximate that function with a polynomial.

Let $g(x) = f(x) - (mx + b)$, where

$$m = \frac{f(1) - f(0)}{1} = f(1) - f(0) \quad \text{and} \quad b = f(0)$$

Since we're adding a polynomial to $f(x)$, $g \in C^0$. Notice that

$$g(0) = f(0) - (f(1) - f(0)) \cdot 0 - f(0) = 0 \quad g(1) = f(1) - f(1) + f(0) - f(0) = 0$$

so $g(0) = 0 = g(1)$. This shows that given any f , it is equivalent to consider f in the "form" of g (the two being homeomorphic), so let $f(0) = 0 = f(1)$. In our construction, we'll sometimes need the domain to extend beyond $[0, 1]$ for technical reasons, so let's say $f(x) = 0$ for $x \in \mathbb{R} - [0, 1]$ (this is a smooth extension with zero measure, so nothing bad can come from this).

Next, we will construct the nice function we'll convolute with. Let

$$\beta_n(t) = b_n(1 - t^2)^n \quad -1 \leq t \leq 1$$

where the constant b_n is chosen so that $\int_{-1}^1 b_n(t)dt = 1$. For $0 \leq x \leq 1$, set

$$P_n(x) = \int_{-1}^1 f(x+t)\beta_n(t)dt$$

This is a weighted average of the value of f using the weight function β_n . This is where we use convolution to get the nice properties of $\beta_n(t)$ to f . We claim that P_n is a polynomial, and $P_n(x) \Rightarrow f(x)$ as $n \rightarrow \infty$.

To check that P_n is a polynomial, we use a change of variables: $u = x + t$. Then for $0 \leq u \leq 1$ we have

$$P_n(x) = \int_{x-1}^{x+1} f(u)\beta_n(u-x)du = \int_0^1 f(u)\beta_n(u-x)du$$

where the last integral is because $f = 0$ outside $[0, 1]$. The function $\beta_n(u-x) = b_n(1 - (u-x)^2)^n$ is a polynomial in x whose coefficients are polynomials in u . We can pull out all the x terms, since the integral is with respect to x , and since the integral is definite with concrete endpoints, it is a constant, and so we have a polynomial!! In other words, we managed to “eliminate” the complexity of f .

To check that $P_n \Rightarrow f$ as $n \rightarrow \infty$, we need to estimate $\beta_n(t)$. We claim that if $\delta > 0$ then

$$\beta_n(t) \Rightarrow 0 \text{ as } n \rightarrow \infty \text{ and } \delta \leq |t| \leq 1$$

Notice that $\delta > 0$, so $0 \leq |t| \leq 1$, and so we are ignoring the fact that $x = 0$ diverges.

You can prove this directly, but I'll skip it for now (p.233)

Now, we'll deduce $P_n \Rightarrow f$. Let $\epsilon > 0$. Since f is uniformly continuous, let $\delta > 0$ such that $|t| < \delta$ implies $|f(x+t) - f(x)| < \frac{\epsilon}{2}$. Since β_n has integral 1 on $[-1, 1]$, we can split in β_n in the following equality:

$$\begin{aligned} |P_n(x) - f(x)| &= \left| \int_{-1}^1 (f(x+t) - f(x)) \beta_n(t) dt \right| \\ &\leq \int_{-1}^1 |f(x+t) - f(x)| \beta_n(t) dt \\ &= \int_{|t| < \delta} |f(x+t) - f(x)| \beta_n(t) dt + \int_{|t| \geq \delta} |f(x+t) - f(x)| \beta_n(t) dt \\ &< \epsilon/2 + 2M \int_{|t| \geq \delta} \beta_n(t) dt \end{aligned}$$

where the 1st integral is less than $\frac{\epsilon}{2}$ because HERE, and the 2nd because HERE. Since δ is positive, and $\beta_n \Rightarrow 0$, then the 2nd integral becomes $\frac{\epsilon}{2}$ as well.

Thus, $P_n \Rightarrow f$ – as we claimed

With this, we can prove the general Stone-Weierstrass theorem on compact metric spaces: that is, change $[0, 1]$ into a general compact metric space.

To do this, we'll first define an algebra on the functions. Let $A \subseteq C^0(M, \mathbb{R})$ such that if $f, g \in A$, $c \in \mathbb{R}$, then

$$f + g \in C^0(M, \mathbb{R}), \quad cf \in C^0(M, \mathbb{R}), \quad f \cdot g \in C^0(M, \mathbb{R})$$

where each operation is defined pointwise (ex. $(f + g)(x) = f(x) + g(x)$). For examples, polynomials form an algebra.

The function algebra *vanishes at point* p if $f(p) = 0$ for all $f \in A$. For example, A with polynomials with zero constant term vanishes at $x = 0$.

The function algebra *separates points* if for each pair of distinct points $p_1, p_2 \in M$, there is a function $f \in A$ such that $f(p_1) \neq f(p_2)$. For example, the function algebra of all trigonometric polynomials separates the points $[0, 2\pi)$ and vanishes nowhere

Theorem 5.6.2: Stone-Weierstrass Theorem

If M is a compact metric space and A is a function-algebra in $C^0(M, \mathbb{R})$ that vanishes nowhere and separates points, then A is dense in $C^0(M, \mathbb{R})$

Note that the Weierstrass theorem is a special case of this one, but it will be used to prove this one

Proof:

We first prove 2 lemmas to help us

If A vanishes nowhere and separates points then there exists $f \in A$ with specified values at any pair of distinct points

Proof.

□

1. power series: definition, integration, differentiation
2. define analytic
3. Weierstrass approximation theorem (set of polynomials is dense in $C^0([a, b], \mathbb{R})$)
4. Stone-Weierstrass Theorem
5. Banach Contraction Principle?
6. Picard's Theorem?

<++>

5.7 Nowhere Differentiable Continuous Functions

Theorem 5.7.1: Nowhere Differentiable Continuous Functions

There exists a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ that has a derivative at no point whatsoever

Proof:

We'll construct a function by taking an infinite series of sawtooth functions which will sum to this:

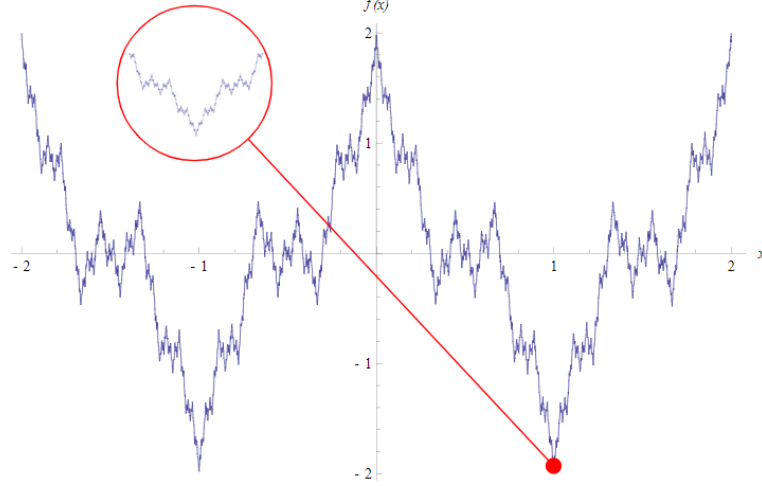


Figure 5.3: Nowhere Differential, yet Continuous Functions

Let the letters k, m, n denote integers. Start with the sawtooth function $\sigma_0 : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$\sigma_0(x) = \begin{cases} x - 2n & \text{if } 2n \leq x \leq 2n + 1 \\ 2n + 2 - x & \text{if } 2n + 1 \leq x \leq 2n + 2 \end{cases}$$

Note that σ_0 is periodic with period 2

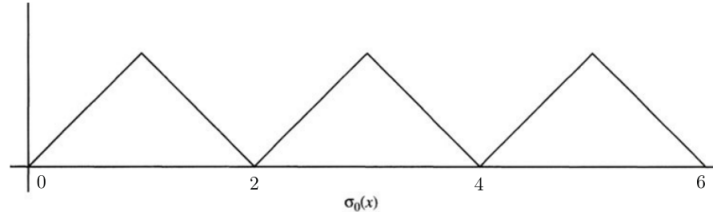


Figure 5.4: σ_0

Next, defined the next sawtooth functions by compression σ_0 :

$$\sigma_k(x) = \left(\frac{3}{4}\right)^k \sigma_0(4^k x)$$

the period will be $\frac{2}{4^k}$, since $\sigma(x) = \sigma(x + m(\frac{2}{4^k}))$ for any $m \in \mathbb{Z}$.

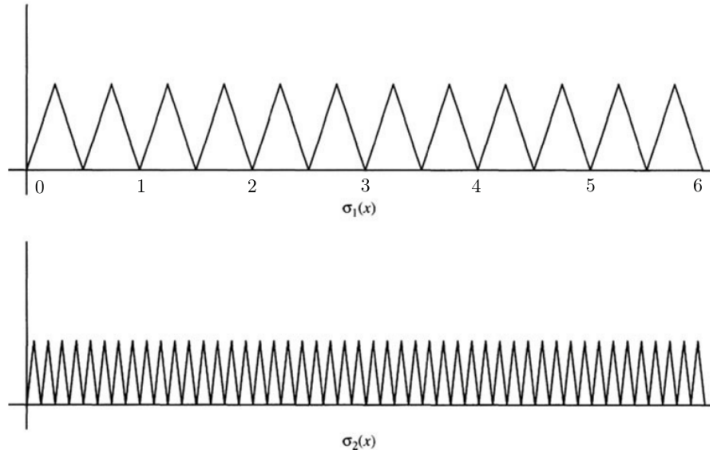


Figure 5.5: two examples of the iteration

By the M -test, $\sum \sigma_k(x)$ converges uniformly to a limit f since $|\sigma_k| \leq \left(\frac{3}{4}\right)^k$ and

$$f(x) = \sum_{k=0}^{\infty} \sigma_k(x)$$

is continuous. We'll show that f is nowhere differentiable, that is

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

does not exist. Since this must work no matter how $h \rightarrow 0$, pick $\delta_m = \frac{1}{2 \cdot 4^m}$. Notice that $\delta_m \rightarrow 0$ as $m \rightarrow \infty$. We'll show that this limit does not exist:

$$\lim_{m \rightarrow 0} \frac{f(x \pm \delta_m) - f(x)}{\delta_m}$$

where the $\pm$ is because we want this limit to represent approach from either the left or right.

Using proposition 5.4.6, we get the derivative is

$$\sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma_k(x)}{\delta_m}$$

We'll split this sum into 3 cases: $k > m$, $k = m$, $k < m$

($k > m$) Since $\delta_m = \frac{2}{4^k}$

$$\delta_m = \frac{1}{2} \frac{1}{4^m} = m \sigma_k \quad m \in \mathbb{Z}$$

and thus

$$\sigma_k(x \pm \delta_m) - \sigma_k(x) = \sigma_k(x) - \sigma_k(x) = 0$$

meaning these terms do not contribute to the sum. We can essentially re-write our sum to

$$\sum_{k=0}^{n-1} \frac{\sigma_k(x \pm \delta_m) - \sigma_k(x)}{\delta_m}$$

($k = m$) σ_m will be monotone on either $[x, x + \delta_m]$ or $[x - \delta_m, x]$:



Figure 5.6: Demonstration of monotonicity

and therefore

$$|\frac{\sigma_m(x + \delta_m) - \sigma_m(x)}{\delta_m}| = 3^m$$

($k < m$) using what we've just learn from the previous exercise, we can crudely estimate each σ_k :

$$\left| \frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m} \right| \leq 3^k$$

giving us

$$\sum_{k < m} \left| \frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m} \right| \leq \sum_{k < m} 3^k = \frac{3^m - 1}{3 - 1}$$

Thus, this gives us

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{m \rightarrow \infty} \sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m} = \lim_{m \rightarrow \infty} 0 \pm 3^m + (3) \geq 3^m - \frac{3^m - 1}{2} = \frac{3^m}{2} + \frac{1}{2}$$

which diverges as $m \rightarrow \infty$! Therefore, this function is *not* differentiable at any point

5.8 Baire Theory

This construction might look like a particular case we need to construct to find, but it turns out that almost all continuous functions are nowhere differentiable! If you define your notion of probability right, then the chance of picking a continuous differentiable function from the set of continuous function is 0%! The way we'll make this rigorous is by taking a detour into Baire spaces.

A set D is dense in M if $\overline{D} = M$. Equivalently, it is dense in M if D for any open $W \subseteq M$, $W \cap D \neq \emptyset$. The intersection of dense sets need not be dense (ex. $\mathbb{Q}$ and $\mathbb{Q}^c$ in $\mathbb{R}$). However, if

U, V are *open-dense* sets in M , then $U \cap V$ will also be open-dense in M . If $W \subseteq M$ is an arbitrary open subset. Then $U \cap W$ is also open, and so by denseness of V , $V \cap (U \cap W)$ is also non-empty! re-arranging using

5.9 Cardinality quick

I'm not sure where to add this yet, but I thought it is a good exercise to appreciate the oddity of $\aleph_1$

Pove that C^0 and $\mathbb{R}$ have the same cardinality

It is clear that C^0 has at least as many function as real numbers since every constant function is continuous, so it comes down to show there are no more. This is exercise 6 in Pugh's book, and its marked with a double star.

Found this too

<https://math.stackexchange.com/questions/477/cardinality-of-set-of-real-continuous-functions>

5.10 Condition for Smooth to be Analytic

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function. Then we can always take its' tailor approximations at a point $T_a f$, and ask whether there exists an open interval U around a such that $T_a f|_U = f|_U$, namely:

$$f(h) = \sum_{k=1}^{\infty} \frac{a_k (h-a)^k}{k!}$$

Not all smooth functions are analytic, as the following typical example demonstrates:

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

which can be verified by taking it's Taylor series which is

$$T_0(f) = \sum_{k=1}^{\infty} 0x^k = 0$$

However, we can precisely understand when a smooth is analytic by looking at the growth rate of the derivative, as we shall demonstrate in this chapter.

We can take a partial Taylor approximation and add on to it the remainder:

$$f(t) = \underbrace{\sum_{j=0}^{k-1} \frac{f^{(j)}(x)}{j!} (t-x)^j}_{\text{Polynomial}} + \underbrace{R_{k-1}(t)}_{\text{Remainder}}$$

Let us estimate $R_k(t)$.

Theorem 5.10.1: Lagrange Remainder Theorem

Let f be a smooth function, or at least $C^{k-1}(\mathbb{R})$ and let U be some interval. Let $P_{k-1}(t)$ be the partial Taylor series around a . Then for fixed t :

$$f(t) = P_{k-1}(t) + \frac{f^{(k)}(\xi_t)}{k!}(t-a)^k$$

Proof:

Given the partial Taylor series $P_{k-1}(t)$ around a , fix a t_0 (without loss of generality $t_0 > a$). Then there always exists a constant C such that:

$$f(t_0) = P_{k-1}(t_0) + \frac{Q}{k!}(t_0 - a)^k$$

Define the function:

$$F(z) = f(t_0) - \sum_{j=0}^{k-1} \frac{f^{(j)}(z)}{j!}(t_0 - z)^j - \frac{Q}{k!}(t_0 - z)^k$$

Then $F(a) = F(t_0) = 0$ and F is differentiable, hence by Rolle's theorem there exists a ξ_t such that:

$$F'(\xi_t) = 0$$

Taking the actual derivative we get:

$$\frac{d}{dz} \left(\frac{f^{(j)}(z)}{j!}(t - z)^j \right) = \frac{f^{(j+1)}(z)}{j!}(t - z)^j - \frac{f^{(j)}(z)}{j!}j(t - z)^{j-1}$$

this creates a telescoping sum gives:

$$F'(z) = -\frac{f^{(k)}(z)}{(k-1)!}(t - z)^{k-1} + Q \frac{k(t - z)^{k-1}}{k!} = \left(-f^{(k)}(z) + Q \right) \frac{(t - z)^{k-1}}{(k-1)!}$$

where the equation is 0 at ξ_t . Now, as $a \neq t_0$, $(t - \xi_t)^{k-1} \neq 0$, hence:

$$(-f^{(k)}(\xi_t) + Q) = 0 \quad \Rightarrow \quad f^{(k)}(\xi_t) = Q$$

as we sought to show.

Using this, we can bound the remainder term. Given a interval $(r - a, r + a) = U$, for any $t \in U$, $|a - t| \leq r$. Then, for any choice of t , we have $f^{(k)}(\xi_t)$, and we can define:

$$M_k = \sup_{t \in U} |f^{(k)}(t)|$$

Hence:

$$|R_{k-1}(t)| = \left| \frac{f^{(k)}(\xi)}{k!}(t - a)^k \right| \leq \frac{M_k}{k!} r^k$$

This is the worst the function can grow. Then by the root test (which compares the growth rate of

of function to $\sum_k \frac{1}{k^n}$ ¹) if the quantity:

$$L = \limsup_{k \rightarrow \infty} \sqrt[k]{\frac{M_k}{k!}} r^k = r \limsup_{k \rightarrow \infty} \sqrt[k]{\frac{M_k}{k!}}$$

is strictly less than 1, it converges, if it is strictly greater than 1 it diverges, and if $L = 1$ then it is indeterminate. Note how R_{k-1} is dependent on the radius r .

Definition 5.10.2: Derivative Growth Rate

Let $I = (x - r, x + r)$ be an interval. Then the *bounded derivative growth rate* of a smooth function f on I is:

$$\alpha_r = \limsup_{k \rightarrow \infty} \sqrt[k]{\frac{M_k}{k!}}$$

Using definition 5.10.2, we have:

$$L = r \cdot \alpha_r$$

Note that α_r is dependent on the radius r as we take the max over an interval of radius r . Now, since $|f^{(k)}(a)| \leq M_k$, we have that

$$R_a = \frac{1}{\limsup_{k \rightarrow \infty} \sqrt[k]{\frac{|f^{(k)}(a)|}{k!}}} \leq \frac{1}{\alpha_r}$$

Hence:

$$\frac{1}{\alpha_r} \leq R_a$$

Theorem 5.10.3: Taylor Convergence

Let I be an interval. If $\alpha \cdot r < 1$ then the Taylor series converges uniformly to f on the interval I .

Proof:

Choose $\delta > 0$ such that $(\alpha + \delta)r < 1$. The Taylor remainder formula from Chapter 3, applied to the $(r - 1)$ st-order remainder, gives

$$f(x + h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!} h^k = \frac{f^{(r)}(\theta)}{r!} h^r$$

for some θ between x and $x + h$. Thus, for r large we have

$$\left| f(x + h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!} h^k \right| \leq \frac{M_r}{r!} r^r = \left(\left(\frac{M_r}{r!} \right)^{1/r} r \right)^r \leq ((\alpha + \delta)r)^r.$$

Since $(\alpha + \delta)r < 1$, the Taylor series converges uniformly to $f(x + h)$ on I .

¹ $L < \rho, 1$, taking n th powers we see by the comparison test that L converges

Theorem 5.10.4: Analytic then bounded derivative growth

If f is expressed as a convergent power series $f(x+h) = \sum c_k h^k$ with radius of convergence $R > \sigma$ then f has bounded derivative growth rate on I .

Proof:

The proof of theorem 5.10.4 uses two estimates about the growth rate of factorials. If you know Stirling's formula they are easy, but we prove them directly.

$$\lim_{r \rightarrow \infty} \frac{r}{\sqrt[r]{r!}} = e \quad (5.1)$$

$$0 < \lambda < 1 \implies \limsup_{r \rightarrow \infty} \sqrt[r]{\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k} < \infty. \quad (5.2)$$

Taking logarithms, applying the integral test, and ignoring terms that tend to zero as $r \rightarrow \infty$ gives

$$\begin{aligned} \frac{1}{r}(\log r! - \log r!) &= \log r - \frac{1}{r}(\log r + \log(r-1) + \cdots + \log 1) \\ &\sim \log r - \frac{1}{r} \int_1^r \log x \, dx = \log r - \frac{1}{r}(x \log x - x) \Big|_1^r \\ &= 1 - \frac{1}{r}, \end{aligned}$$

which tends to 1 as $r \rightarrow \infty$. This proves (5.1).

To prove (5.2) we write $\lambda = e^{-\mu}$ for $\mu > 0$, and reason similarly:

$$\begin{aligned} \sum_{k=r}^{\infty} \binom{k}{r} \lambda^k &= \sum_{k=r}^{\infty} \frac{k(k-1)\cdots(k-r+1)}{r!} e^{-k\mu} \\ &\leq \frac{1}{r!} \sum_{k=r}^{\infty} k^r e^{-k\mu} \approx \frac{1}{r!} \int_r^{\infty} x^r e^{-\mu x} \, dx \\ &= \frac{-1}{r!} e^{-\mu x} \left(\frac{x^r}{\mu} + \frac{rx^{r-1}}{\mu^2} + \frac{r(r-1)x^{r-2}}{\mu^3} + \cdots + \frac{r!}{\mu^{r+1}} \right) \Big|_r^{\infty} \\ &\leq \frac{1}{r!} e^{-\mu r} (r+1)r^r \frac{1}{\min(1, \mu)^{r+1}}. \end{aligned}$$

According to (5.1) the r th root of this quantity tends to $e^{1-\mu}/\min(1, \mu)$ as $r \rightarrow \infty$, completing the proof of (5.2).

By assumption the power series $\sum c_k h^k$ has radius of convergence R and $\sigma < R$. Since $1/R$ is the lim sup of $\sqrt[k]{|c_k|}$ as $k \rightarrow \infty$, there is a number $\lambda < 1$ such that for all large k we have $|c_k \sigma^k| \leq \lambda^k$. Differentiating the series term by term with $|h| \leq \sigma$ gives

$$\begin{aligned} |f^{(r)}(x+h)| &\leq \sum_{k=r}^{\infty} k(k-1)(k-2)\cdots(k-r+1) |c_k h^{k-r}| \\ &\leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} |c_k \sigma^k| \leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} \lambda^k. \end{aligned}$$

for r large. Thus,

$$M_r = \sup_{|h| \leq \sigma} |f^{(r)}(x+h)| \leq \frac{r!}{\sigma^r} \sum_{k=r}^{\infty} \binom{k}{r} \lambda^k.$$

According to (5.2),

$$\alpha = \limsup_{r \rightarrow \infty} \sqrt[r]{\frac{M_r}{r!}} \leq \frac{1}{\sigma} \limsup_{r \rightarrow \infty} \sqrt[r]{\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k} < \infty,$$

and f has bounded derivative growth rate on I .

Theorem 5.10.5: Analyticity Theorem

A smooth function is analytic if and only if it has locally bounded derivative growth rate.

Proof:

Assume that $f : (a, b) \rightarrow \mathbb{R}$ is smooth and has locally bounded derivative growth rate. Then $x \in (a, b)$ has a neighborhood N on which the derivative growth rate α is finite. Choose $\sigma > 0$ such that $I = [x - \sigma, x + \sigma] \subset N$ and $\alpha\sigma < 1$. We infer from theorem 5.10.3 that the Taylor series for f at x converges uniformly to f on I . Hence f is analytic.

Conversely, assume that f is analytic and let $x \in (a, b)$ be given. There is a power series $\sum c_k h^k$ that converges to $f(x+h)$ for all h in some interval $(-R, R)$ with $R > 0$. Choose σ with $0 < \sigma < R$. We infer from theorem 5.10.4 that f has bounded derivative growth rate on I .

Thus, while smooth functions have no bounds for the derivative:

$$|f^{(k)}(x)| \leq \text{anything, no bound}$$

analytic functions satisfy:

$$|f^{(k)}(x)| \leq C \cdot \frac{k!}{R^k}$$

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